Project Report

Pollen’s Profiling – Automated Classification of Pollen Grains

Team ID: LTVIP2025TMID45577

Team Leader: Pooja Harshitha

Team Members:

- Navya Ramisetty
- Narepalem Teja
- Naragam Manikanta Raghava

1. Introduction

Pollen’s Profiling is a deep learning project aimed at automating the classification of pollen grain images. This task, traditionally dependent on expert observation through microscopes, is time-consuming and prone to human error. Our solution leverages Convolutional Neural Networks (CNNs) to classify pollen into 23 distinct classes with considerable accuracy. This tool benefits researchers, allergists, and agricultural scientists who require rapid and reliable identification of pollen species.

2. Ideation Phase

2.1 Problem Statement

Manual pollen grain classification is slow, inconsistent, and resource-intensive. An AI-based solution can bring speed, consistency, and accessibility to the process.

2.2 Empathy Mapping

User	Says	Thinks	Does	Feels
Researcher	"It takes hours to sort images."	"There must be a better way."	Observes under microscope	Frustrated

User	Says	Thinks	Does	Feels
Student	"How do I identify this?"	"This is too advanced for me."	Searches online guides	Confused
Lab Assistant	"Results differ every time."	"Manual methods are unreliable."	Notes down guesses	Insecure

2.3 Brainstorming Ideas

- Use CNN for visual pattern recognition.
- Build lightweight Flask web app.
- Handle class imbalance with weighted loss.
- Enable image upload + instant prediction.

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3. Requirement Analysis

3.1 Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Image Upload Interface	Upload image via HTML form
FR-2	Classification & Prediction	Classify image using CNN
		Return predicted pollen class
		Display confidence score
FR-3	Result Output	Show predicted result on web page
FR-4	Data Logging (Future Scope)	Log image metadata and predictions
FR-5	Admin Monitoring	View all predictions history
FR-6	Report Export (Future Scope)	Allow exporting of prediction logs as CSV

3.2 Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Interface should be clean and usable by students/researchers
NFR-2	Security	Input validation, restricted file types, server-side filtering
NFR-3	Reliability	Model should give consistent outputs for the same image
NFR-4	Performance	Classification should occur within 2–4 seconds
NFR-5	Availability	System should run 99.9% of the time during demo/deployment
NFR-6	Scalability	Should allow for expansion to more classes/images in the future

4. Project Design

4.1 Problem–Solution Fit

#	Component	Description
1	Customer Segment	Researchers, students, allergists
2	Jobs-to-be-Done	Classify 23 pollen types accurately and quickly
3	Triggers	Needing fast classification of microscope images
4	Emotions	Before: Frustrated. After: Confident
5	Available Solutions	Manual microscopy, paper guides
6	Customer Constraints	Image clarity, class imbalance, hardware limitations
7	Behavior	Uploads images and uses predictions in reports or analysis
8	Channels	Flask-based lightweight web app
9	Problem Root Cause	Manual sorting is slow, inconsistent, and unscalable
10	Our Solution	CNN + Flask, 790 training images, real-time web predictions

4.2 Proposed Solution

A Flask web app that allows users to upload pollen images which are then classified using a CNN model trained on 23 classes. Predictions and confidence levels are displayed instantly.

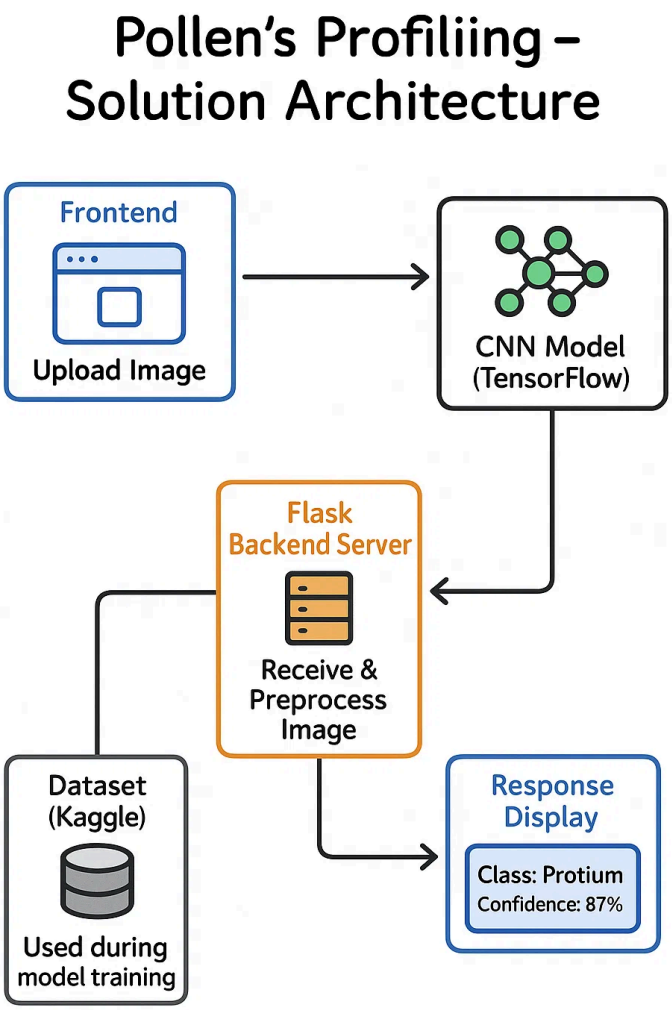
📄 Proposed Solution Details

S.No.	Parameter	Description
1	**Problem Statement**	Manual identification of pollen grains is time-consuming, error-prone, and requires expert knowledge; there is a need for an automated, scalable solution for accurate classification.
2	**Idea / Solution Description**	A CNN-powered image classification system deployed via a Flask web application that allows users to upload pollen images and instantly receive species identification among 23 classes.
3	**Novelty / Uniqueness**	Leverages data augmentation and transfer learning on a specialized Brazilian Savannah Pollen Dataset to achieve high accuracy with limited samples; offers real-time inference on edge devices.
4	**Social Impact / Customer Satisfaction**	Accelerates environmental and agricultural research, aids allergists in pollen allergy diagnosis, and educates users on local pollen distributions—improving both scientific outcomes and public health.
5	**Business Model (Revenue Model)**	Subscription-based SaaS for research institutions and allergy clinics; licensable API for agricultural companies; freemium web portal for educational use by students and hobbyists.
6	**Scalability of the Solution**	Easily extensible to new pollen classes via fine-tuning; containerized Flask deployment supports cloud, on-prem, and offline edge deployment for field researchers and remote labs.

🔗 Reference

Title	Details
Reference Paper	*Deep learning for accurate classification of conifer pollen grains: enhancing species identification in palynology*
Description	Discusses the application of deep learning and transfer learning for conifer pollen classification, highlighting challenges and accuracy improvements in palynology research.
Link	https://www.frontiersin.org/journals/big-data/articles/10.3389/fdata.2025.1507036/full

4.3 Architecture Diagram



5. Project Planning & Scheduling

5.1 Product Backlog and Sprint Allocation

User Type	Epic	Story No.	Task	Sprint
User	Upload Image	USN-1	Create HTML form	Sprint-1
User	Prediction Display	USN-2	Return & show prediction	Sprint-1
Admin	Log Monitoring	USN-3	View prediction logs	Sprint-2
System	Run CNN Model	USN-4	Inference pipeline	Sprint-1
System	Save Metadata	USN-5	Log result info	Sprint-2
User	Report Export	USN-6	CSV export (future scope)	Sprint-3

5.2 Tracker

Maintained in Google Sheets with weekly check-ins and velocity tracking.

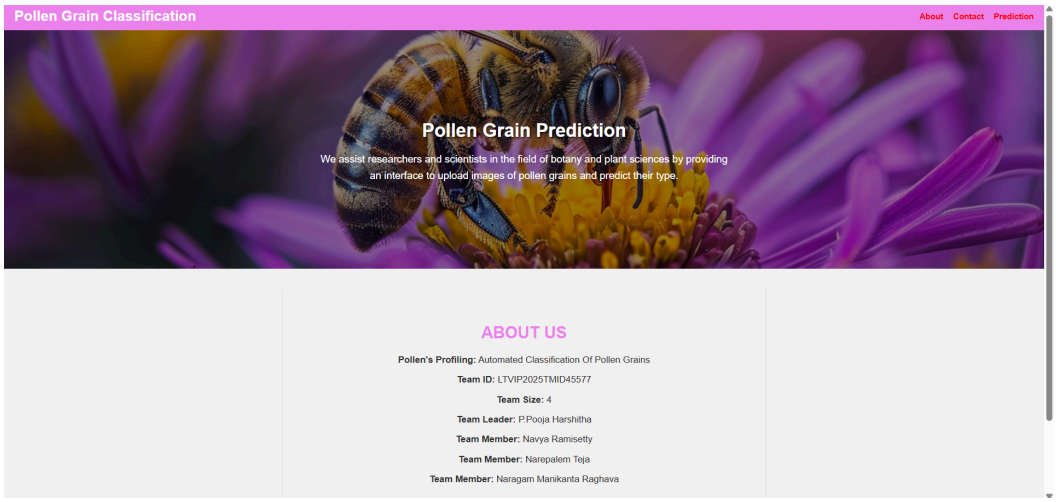
5.3 Burn Down Chart

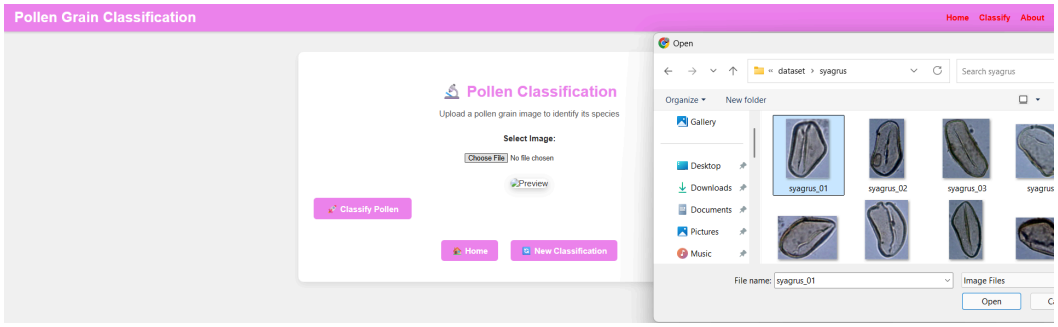
Visualized based on sprint completion — used to monitor progress and task completion rate.

6. Functional and Performance Testing

- **Test Accuracy:** 87.6%
- **Loss Value:** ~0.707
- **Model:** CNN with weighted loss for imbalanced dataset
- **Evaluation Tools:** Classification report, confusion matrix
- **Average Inference Time:** ~2.8 seconds

7. Results





8. Advantages & Disadvantages

Advantages	Disadvantages
Fast and scalable	Needs high-quality images

Advantages	Disadvantages
User-friendly interface	Dataset imbalance affects rare classes
Model explainability via score	Requires web deployment setup

9. Conclusion and Future Scope

This project demonstrates the effective use of deep learning in ecological and biological classification problems. By automating pollen classification, we reduce human dependency and improve consistency. Future plans include:

- Mobile integration
- Dataset expansion
- Real-time microscope image feeds
- Transfer learning upgrades

10. Appendix

- **GitHub Repository:**

  https://github.com/NMRL24/pollens-profiling-automated-classification-of-pollen-grains

- **Demo Video:**

  https://drive.google.com/file/d/1kN26OI_PGkHjNmTMIRHIAcHumIcSNfup/view?usp=sharing

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Pollen Grain Classification</title>
  <style>
   body {
```

```
  font-family: Arial, sans-serif;
  margin: 0;
  padding: 0;
  background-color: #f4f4f4;
  color: #333;
}

header {
  background-color: #EE82EE;
  padding: 10px 20px;
  color: white;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  box-shadow: 0 2px 5px rgba(0,0,0,0.2);
}

header h1 {
  font-size: 28px;
  font-weight: bold;
  margin: 0;
}

nav a {
  color: red;
  margin-left: 15px;
  text-decoration: none;
  font-weight: bold;
  position: relative;
  transition: color 0.3s ease;
}

nav a::after {
  content: '';
  position: absolute;
```

```
  width: 0%;
  height: 2px;
  bottom: -2px;
  left: 0;
  background-color: red;
  transition: width 0.3s ease;
}

nav a:hover {
  color: white;
}

nav a:hover::after {
  width: 100%;
}

.hero {
  background-image: linear-gradient(rgba(0,0,0,0.4), rgba(0,0,0,0.4)), url("{{ ur
  background-size: cover;
  background-position: center;
  color: white;
  text-align: center;
  padding: 145px 100px;
}

.hero h2 {
  font-size: 36px;
  font-weight: bold;
  margin-bottom: 20px;
  text-shadow: 2px 2px 4px rgba(0,0,0,0.7);
}

.hero p {
  font-size: 20px;
  margin-top: 20px;
  line-height: 1.6;
```

```
  max-width: 800px;
  margin-left: auto;
  margin-right: auto;
}

.about {
  text-align: center;
  padding: 50px 20px;
  background-color: #f2f2f2;
  color: #333;
  margin: 30px auto;
  max-width: 900px;
  border-radius: 8px;
  box-shadow: 0 4px 8px rgba(0,0,0,0.1);
}

.about h2 {
  font-size: 32px;
  margin-bottom: 20px;
  color: #EE82EE;
}

.about p {
  max-width: 800px;
  margin: 0 auto 10px auto;
  font-size: 18px;
  line-height: 1.6;
  text-align: center;
}

footer {
  background-color: #333;
  color: white;
  text-align: center;
  padding: 20px;
  margin-top: 50px;
```

```
    font-size: 14px;
  }

  @media (max-width: 768px) {
    .hero h2 {
      font-size: 28px;
    }

    .hero p {
      font-size: 16px;
    }

    nav a {
      display: block;
      margin: 10px 0;
    }

    header {
      flex-direction: column;
      align-items: flex-start;
    }
  }
 </style>
</head>
<body>

 <header>
  <h1>Pollen Grain Classification</h1>
  <nav>
   <a href="/">About</a>
   <a href="#">Contact</a>
   <a href="/predict">Prediction</a>
  </nav>
 </header>

 <main>
```

```
    <section class="hero">
      <h2>Pollen Grain Prediction</h2>
      <p>
        We assist researchers and scientists in the field of botany and plant sciences
      </p>
    </section>

    <section class="about" id="about">
      <h2>ABOUT US</h2>
      <p><strong>Pollen's Profiling:</strong> Automated Classification Of Pollen G
      <p><strong>Team ID:</strong> LTVIP2025TMID45577</p>
      <p><strong>Team Size:</strong> 4</p>
      <p><strong>Team Leader:</strong> P.Pooja Harshitha</p>
      <p><strong>Team Member:</strong> Navya Ramisetty</p>
      <p><strong>Team Member:</strong> Narepalem Teja</p>
      <p><strong>Team Member:</strong> Naragam Manikanta Raghava</p>
    </section>
  </main>

  <footer>
    <p>&copy; 2025 Pollen Grain Classification. All rights reserved.</p>
  </footer>

</body>
</html>
```

predict.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Pollen Classification - Predict</title>
  <style>
    body {
```

```
  font-family: Arial, sans-serif;
  margin: 0;
  padding: 0;
  background-color: #f4f4f4;
  color: #333;
}

header {
  background-color: #EE82EE;
  padding: 10px 20px;
  color: white;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  box-shadow: 0 2px 5px rgba(0,0,0,0.2);
}

header h1 {
  font-size: 28px;
  font-weight: bold;
  margin: 0;
}

nav a {
  color: red;
  margin-left: 15px;
  text-decoration: none;
  font-weight: bold;
  position: relative;
  transition: color 0.3s ease;
}

nav a::after {
  content: '';
  position: absolute;
```

```
  width: 0%;
  height: 2px;
  bottom: -2px;
  left: 0;
  background-color: red;
  transition: width 0.3s ease;
}

nav a:hover {
  color: white;
}

nav a:hover::after {
  width: 100%;
}

.container {
  max-width: 900px;
  margin: 40px auto;
  padding: 0 20px;
}

.main-card {
  background-color: white;
  border-radius: 10px;
  padding: 30px;
  box-shadow: 0 5px 15px rgba(0,0,0,0.1);
}

.card-header {
  text-align: center;
  margin-bottom: 30px;
}

.card-header h2 {
  font-size: 30px;
```

```
  color: #EE82EE;
  margin-bottom: 10px;
}

.card-header p {
  font-size: 16px;
  color: #555;
}

.file-upload-wrapper {
  margin-bottom: 20px;
  text-align: center;
}

input[type="file"] {
  padding: 10px;
  margin-top: 10px;
}

.file-name {
  font-style: italic;
  font-size: 14px;
  color: #555;
  margin-top: 5px;
}

.image-preview img {
  max-width: 100%;
  max-height: 300px;
  margin-top: 15px;
  border-radius: 10px;
  box-shadow: 0 4px 12px rgba(0,0,0,0.2);
}

.submit-button {
  background-color: #EE82EE;
```

```
  color: white;
  border: none;
  padding: 10px 25px;
  font-size: 16px;
  font-weight: bold;
  border-radius: 6px;
  cursor: pointer;
  transition: background 0.3s ease;
}

.submit-button:hover {
  background-color: #d76add;
}

.loading {
  text-align: center;
  margin-top: 15px;
}

.result-section {
  margin-top: 30px;
  background-color: #f2f2f2;
  padding: 20px;
  border-radius: 10px;
  text-align: center;
}

.species {
  font-size: 24px;
  font-weight: bold;
  color: #4CAF50;
}

.confidence {
  font-size: 16px;
  margin-top: 10px;
```

```
    }

    .nav-links {
      text-align: center;
      margin-top: 30px;
    }

    .nav-links a {
      display: inline-block;
      margin: 10px;
      padding: 10px 20px;
      background-color: #EE82EE;
      color: white;
      text-decoration: none;
      border-radius: 6px;
      font-weight: bold;
      transition: background 0.3s ease;
    }

    .nav-links a:hover {
      background-color: #d76add;
    }
  </style>
</head>
<body>

  <header>
    <h1>Pollen Grain Classification</h1>
    <nav>
      <a href="/">Home</a>
      <a href="/predict">Classify</a>
      <a href="/#about">About</a>
    </nav>
  </header>

  <div class="container">
```

```
<div class="main-card">
  <div class="card-header">
    <h2>🔬 Pollen Classification</h2>
    <p>Upload a pollen grain image to identify its species</p>
  </div>

  <form method="POST" enctype="multipart/form-data" id="uploadForm">
    <div class="file-upload-wrapper">
      <label for="fileInput"><strong>Select Image:</strong></label><br />
      <input type="file" name="image" accept="image/*" required id="fileInput"
      <div class="file-name" id="fileName"></div>
      <div class="image-preview" id="imagePreview">
        <img id="previewImg" src="" alt="Preview">
      </div>
    </div>
    <button type="submit" class="submit-button" id="submitBtn" disabled>🚀 C
    <div class="loading" id="loading" style="display: none;">
      <p>Analyzing image...</p>
    </div>
  </form>

  {% if prediction and image_path %}
  <div class="result-section" id="resultSection">
    <div class="species">🌿 {{ prediction }}</div>
    {% if confidence %}
    <div class="confidence">Confidence: {{ '%.2f' % confidence }}%</div>
    {% else %}
    <div class="confidence">Confidence: Not Available</div>
    {% endif %}
    <div style="margin-top: 15px;">
      <img src="{{ url_for('static', filename='uploads/' + image_path) }}" alt="Up
    </div>
  </div>
  {% endif %}

  <div class="nav-links">
```

```
      <a href="/">🏠 Home</a>
      <a href="/predict">🔄 New Classification</a>
    </div>
  </div>
</div>

<script>
  const fileInput = document.getElementById('fileInput');
  const fileName = document.getElementById('fileName');
  const imagePreview = document.getElementById('imagePreview');
  const previewImg = document.getElementById('previewImg');
  const submitBtn = document.getElementById('submitBtn');
  const loading = document.getElementById('loading');

  fileInput.addEventListener('change', function (e) {
    const file = e.target.files[0];
    if (file) {
      fileName.textContent = `Selected: ${file.name}`;
      fileName.style.display = 'block';
      const reader = new FileReader();
      reader.onload = function (e) {
        previewImg.src = e.target.result;
        imagePreview.style.display = 'block';
      };
      reader.readAsDataURL(file);
      submitBtn.disabled = false;
    } else {
      fileName.style.display = 'none';
      imagePreview.style.display = 'none';
      submitBtn.disabled = true;
    }
  });

  document.getElementById('uploadForm').addEventListener('submit', function (
    if (!fileInput.files.length) {
      e.preventDefault();
```

```
      alert('Please select an image.');
    } else {
      submitBtn.style.display = 'none';
      loading.style.display = 'block';
    }
  });
 </script>
</body>
</html>
```

logout.html

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Logout</title>
 <style>
  body {
    margin: 0;
    font-family: Arial, sans-serif;
    background-color: #f4f4f4;
    text-align: center;
    color: #333;
  }

  header {
    background-color: #EE82EE;
    padding: 10px 20px;
    color: white;
    display: flex;
    justify-content: space-between;
    align-items: center;
    box-shadow: 0 2px 5px rgba(0,0,0,0.2);
  }
```

```
nav a {
  color: red;
  margin-left: 15px;
  text-decoration: none;
  font-weight: bold;
  position: relative;
}

nav a::after {
  content: '';
  position: absolute;
  width: 0%;
  height: 2px;
  bottom: -2px;
  left: 0;
  background-color: red;
  transition: width 0.3s ease;
}

nav a:hover {
  color: white;
}

nav a:hover::after {
  width: 100%;
}

.logout-section {
  background-image: linear-gradient(rgba(0,0,0,0.4), rgba(0,0,0,0.4)),
                    url("{{ url_for('static', filename='xbee.jpg') }}");
  background-size: cover;
  background-position: center;
  color: white;
  padding: 120px 20px;
  min-height: 500px;
  display: flex;
```

```
    flex-direction: column;
    justify-content: center;
    align-items: center;
  }

  .logout-section h2 {
    font-size: 36px;
    margin-bottom: 20px;
    text-shadow: 2px 2px 4px rgba(0,0,0,0.7);
  }

  .logout-section a {
    background-color: #007C80;
    color: white;
    padding: 10px 25px;
    font-size: 16px;
    border-radius: 6px;
    text-decoration: none;
    font-weight: bold;
    box-shadow: 0 0 8px rgba(0,0,0,0.3);
  }

  .logout-section a:hover {
    background-color: #005f61;
  }

  footer {
    background-color: #333;
    color: white;
    text-align: center;
    padding: 20px;
    font-size: 14px;
  }
 </style>
</head>
<body>
```

```
  <header>
    <h1>Pollen Grain Classification</h1>
    <nav>
      <a href="/">About</a>
      <a href="#">Contact</a>
      <a href="/predict">Prediction</a>
    </nav>
  </header>

  <div class="logout-section">
    <h2>You have successfully logged out.</h2>
    <a href="/">Back to Home</a>
  </div>

  <footer>
    <p>&copy; 2025 Pollen Grain Classification. All rights reserved.</p>
  </footer>

</body>
</html>
```

app.py

```
import os
import numpy as np
from flask import Flask, request, render_template
from keras.preprocessing import image
from keras.models import load_model

# Initialize Flask app
app = Flask(__name__)

# Load your trained Keras model
model = load_model("model.h5", compile=False)
```

```
# List of class labels (ensure it matches your model's output order)
labels = [
    'anadenanthera', 'arecaceae', 'arrabidaea', 'cecropia', 'chromolaena',
    'combretum', 'croton', 'dipteryx', 'eucalipto', 'faramea', 'hyptis',
    'mabea', 'matayba', 'mimosa', 'myrcia', 'protium', 'qualea', 'schinus',
    'senegalia', 'serjania', 'syagrus', 'tridax', 'urochloa'
]

# Home route
@app.route('/')
@app.route('/index.html')
def index():
    return render_template('index.html')

# Prediction route
@app.route('/predict', methods=['GET', 'POST'])
def predict():
    if request.method == 'POST':
        file = request.files['image']
        if not file:
            return render_template('predict.html', prediction=None, image_path=None

        # Save uploaded file to static/uploads directory
        basepath = os.path.dirname(__file__)
        upload_folder = os.path.join(basepath, 'static', 'uploads')
        os.makedirs(upload_folder, exist_ok=True)
        upload_path = os.path.join(upload_folder, file.filename)
        file.save(upload_path)

        # Preprocess the image
        img = image.load_img(upload_path, target_size=(128, 128))
        x = image.img_to_array(img)
        x = np.expand_dims(x, axis=0)

        # Predict
        probs = model.predict(x)[0]
```

```
        pred = np.argmax(probs)
        predicted_label = labels[pred]
        confidence = float(probs[pred] * 100)

        return render_template('predict.html',
                               prediction=predicted_label,
                               image_path=file.filename,
                               confidence=confidence)

    return render_template('predict.html', prediction=None, image_path=None, cor

# Logout route
@app.route('/logout.html')
def logout():
    return render_template('logout.html')

# Run the app
if __name__ == "__main__":
    app.run(debug=True)
```